



Python Dictionary

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Python dictionary is a data structure that stores information in key-value pairs. While keys must be unique and immutable (like strings or numbers), values can be of any data type, whether mutable or immutable. This makes dictionaries ideal for accessing data by a specific name rather than a numeric position like in list.

Example: This example shows how a dictionary stores data using keys and values.

```
data = { "name": "Jake", "age": 22 }  
print(data)
```



Output

```
{'name': 'Jake', 'age': 22}
```

Explanation: "name" and "age" are keys, "Jake" and 22 are their values and dictionary stores data in key:value format

Creating a Dictionary

A dictionary is created by writing key-value pairs inside { }, where each key is connected to a value using colon (:). A dictionary can also be created using [dict\(\)](#) function.

```
a = {1: 'Geeks', 2: 'For', 3: 'Geeks'}  
print(a)  
  
b = dict(a = "Geeks", b = "for", c = "Geeks")  
print(b)
```



Output

```
{1: 'Geeks', 2: 'For', 3: 'Geeks'}  
{'a': 'Geeks', 'b': 'for', 'c': 'Geeks'}
```

Accessing Dictionary Items

A value in a dictionary is accessed by using its key. This can be done either with square brackets `[]` or with the `get()` method. Both return the value linked to the given key.

```
d = { "name": "Kat", 1: "Python", (1, 2): [1,2,4] }

# Access using key
print(d["name"])

# Access using get()
print(d.get("name"))
```

Output

```
Kat
Kat
```

Note: Accessing a missing key with `[]` raises a `KeyError`, while `get()` is safer because it returns `None` (or a default value) instead of an error.

Adding and Updating Dictionary Items

New items are added to a dictionary using the assignment operator (`=`) by giving a new key a value. If an existing key is used with the assignment operator, its value is updated with the new one.

```
d = {1: 'Geeks', 2: 'For', 3: 'Geeks'}

# Adding a new key-value pair
d["age"] = 22

# Updating an existing value
d[1] = "Python dict"
print(d)
```

Output

```
{1: 'Python dict', 2: 'For', 3: 'Geeks', 'age': 22}
```

Removing Dictionary Items

Dictionary items can be removed using built-in deletion methods that work on keys:

- `del`: removes an item using its key
- `pop()`: removes the item with the given key and returns its value
- `clear()`: removes all items from the dictionary
- `popitem()`: removes and returns the last inserted key-value pair

```
d = {1: 'Geeks', 2: 'For', 3: 'Geeks', 'age': 22}

# Using del
del d["age"]
print(d)

# Using pop()
val = d.pop(1)
print(val)
```

```
# Using popitem()
key, val = d.popitem()
print(f"Key: {key}, Value: {val}")

# Using clear()
d.clear()
print(d)
```

Output

```
{1: 'Geeks', 2: 'For', 3: 'Geeks'}
Geeks
Key: 3, Value: Geeks
{}
```

Iterating Through a Dictionary

A dictionary can be traversed using a for loop to access its keys, values or both key-value pairs by using the built-in methods [keys\(\)](#), [values\(\)](#), and [items\(\)](#).

```
d = {1: 'Geeks', 2: 'For', 'age':22}

# Iterate over keys
for key in d:
    print(key)

# Iterate over values
for value in d.values():
    print(value)

# Iterate over key-value pairs
for key, value in d.items():
    print(f"{key}: {value}")
```

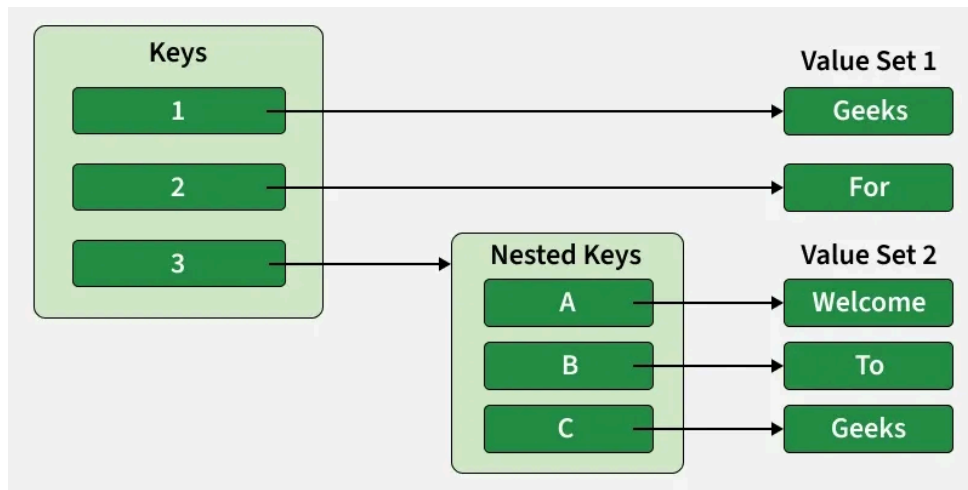
Output

```
1
2
age
Geeks
For
22
1: Geeks
2: For
age: 22
```

Read in detail: [Ways to Iterating Over a Dictionary](#)

Nested Dictionaries

A nested dictionary is a dictionary that contains another dictionary as one of its values. Below diagram shows how a nested dictionary works, where key 3 points to another dictionary inside the main dictionary.



Representation of Nested Dictionary

The arrows show how each key is connected to its corresponding value.

```
d = {1: 'Geeks', 2: 'For', 3: {'A': 'Welcome', 'B': 'To', 'C': 'Geeks'}}  
print(d)
```

Output

```
{1: 'Geeks', 2: 'For', 3: {'A': 'Welcome', 'B': 'To', 'C': 'Geeks'}}
```

Read in Detail: [Python Nested Dictionary](#)

Recommended Problems: [Intro to Dictionary](#), [Dictionary II](#), [Dictionary III](#), [Uncommon characters](#), [Dictionary Practice Problems](#)

Related Links:

- [Dictionary Comprehension](#)
- [Dictionary Methods](#)
- [Dictionary Copy Vs Deep Copy](#)

Suggested Quiz

6 Questions

What is the output of the following code?

```
d = {"a": 1, "b": 2}
print(d["a"])
```

☐ A a☐ B 1☐ C 2[Login to View Explanation](#)

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